

DSA0502 – Query Processing for Data Science	
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LAB PROGRAMS 21 AND 22

21. Swap the Cases of a Specified Character Column

Question

Write a Pandas program to swap the cases of a specified character column in a given DataFrame.

Aim

To convert uppercase letters into lowercase and lowercase letters into uppercase in a DataFrame column.

Dataset — names.csv

```
%%writefile names.csv
```

Name

Ravi

PRIYA

ArJuN

mEeNa

Python Program

```
import pandas as pd

# Read the dataset
df = pd.read_csv("names.csv")
print("Original Dataset:")
print(df)

# Swap the cases
df["Name"] = df["Name"].str.swapcase()
print("\nAfter Swapping Cases:")
print(df)
```

Output

```

... Original Dataset:
      Name
0    Ravi
1  PRIYA
2  ArJuN
3  mEeNa

      After Swapping Cases:
      Name
0    rAVI
1  priya
2  aRjUn
3  MeEnA

```

Result

The uppercase and lowercase characters in the specified column were swapped successfully.

22. Draw a Line with Labels and Title

Question

Write a Python program to draw a line with suitable labels on the x-axis, y-axis, and a title.

Aim

To draw a simple line graph with suitable x-axis label, y-axis label, and chart title.

Dataset — line_data.csv

```
%%writefile line_data.csv
```

```
X,Y
```

```
1,2
```

```
2,4
```

```
3,6
```

```
4,8
```

```
5,10
```

Python Program

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
# Read the dataset
```

```
df = pd.read_csv("line_data.csv")
```

```
print("Dataset:")
```

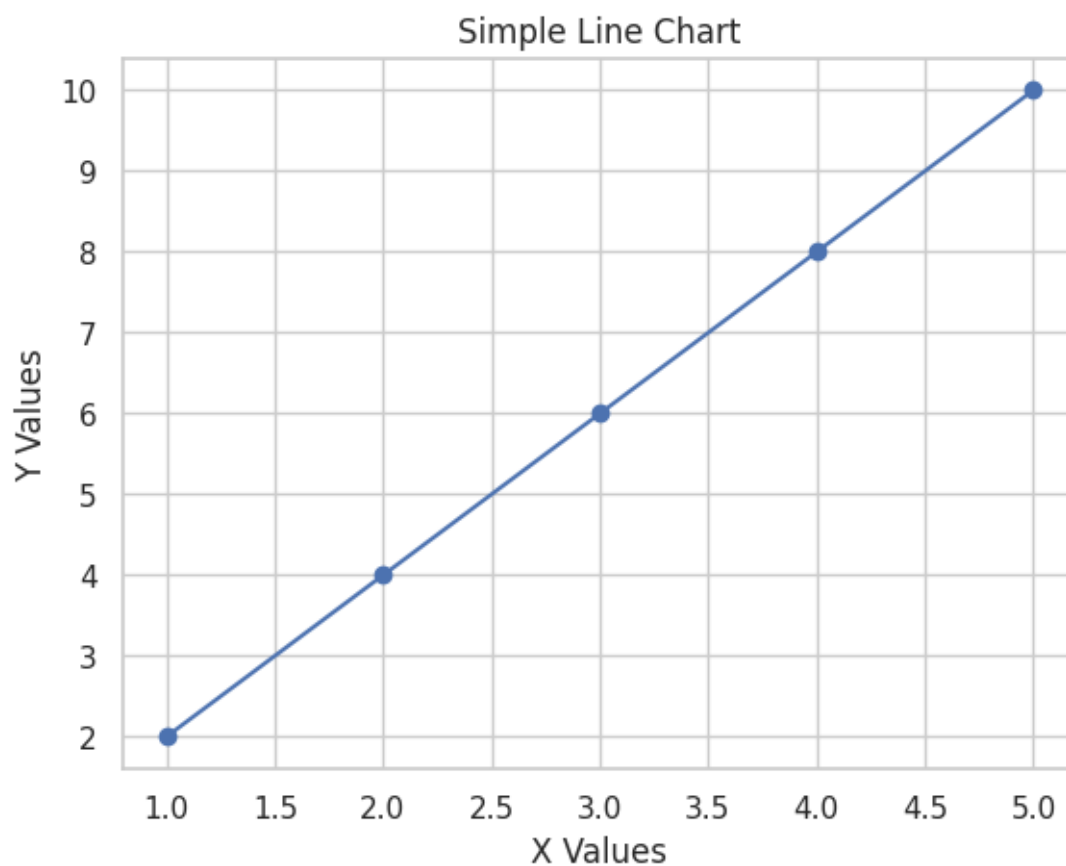
```
print(df)
```

```
# Draw the line chart
```

```
plt.plot(df["X"], df["Y"], marker="o")  
# Add labels and title  
plt.xlabel("X Values")  
plt.ylabel("Y Values")  
plt.title("Simple Line Chart")  
# Display the graph  
plt.show()
```

Output

```
... Dataset:  
   X  Y  
0  1  2  
1  2  4  
2  3  6  
3  4  8  
4  5 10
```



Result

A line chart with suitable axis labels and title was created successfully.